

FOST TRAINING RULE OF THE ROAD TEST APR 16

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(F) Vol I and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90% (45 out of 50)

You are the OOW in HMS ST ALBANS (length 133m) deploying East of Suez for OP KIPION conducting exercises enroute. You have just sailed from Gibraltar to commence a NATO exercise in the Mediterranean. It is the forenoon watch and SSD are falling out as you take the ship from the NO. Visibility is good but fog is forecast for later in the day. You sight at R40 a sea plane making an approach to land on the sea surface. You ascertain that the range is reducing and the bearing is steady.

1. Do the Rules apply to the Sea Plane? If so, how should the sea plane operate, once it has landed? (R.3, R.18e) (3)

2. As you assess there is a risk of collision, does this change any obligations between you and the Sea Plane? Explain your answer (R.8fiii) (2)

The Sea Plane passes at a safe distance, and you continue on passage towards Toulon, to meet with the rest of the exercise group. As predicted the visibility starts to reduce you assess visibility currently as 8nm.

3. In determining a safe speed what factors shall be among those taken into account?

a. By all vessels: (R6) (3)

b. And by vessels with operational radar? (R6) (3)

4. You have received a Navtex message indicting that there is a 60m tug towing a 200m barge ivo Europa point. You see a double echo on the radar at a range of 8 miles. What light configuration would you expect to see on that bearing if it was the tug and tow coming towards you? (R24, Annex I 2 e) (4)

5. You continue east and you sight visually a container ship on the ships head on a reciprocal course. Who should give way? (R. 14a) (1)

6. What do the Rules state about if you have any doubt as to this situation? (R. 14c) (1)

7. How large an alteration should be made to avoid collision? (R. 8b) (1)

8. You notice you are slowly overhauling a small merchant vessel on a similar track to your own. When shall you be deemed to be overtaking? (R. 13b) (2)

9. Who should keep out of the way? (R. 13a) (1)

With the shipping situation resolved you and the YO commence some rule of the road revision.

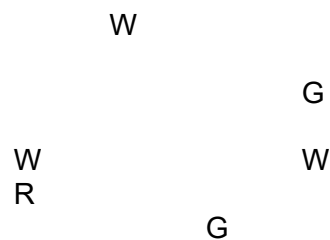
10. In construing and complying with these rules due regard shall be had to... (R.2b) (2)

11. Define the term "power-driven vessel" (R.3b) (1)

12. Define the term "wing in ground craft" (R.3m) (2)

The visibility continues to reduce and you now assess it to be 4nm.

13. You sight the below lights at Green 45 amongst a collection of similar lights.



a. What is this vessel? (R.26 & Annex II) (3)

b. Would any of the vessels lights change if it was not making way through the water? (R.26) (1)

14. A vessel is sighted at RED 50, range 4 miles just visual coming out of the fog bank. The range is closing and the bearing is moving very slowly right. You are 40° on his starboard bow. What action is required by the Rules? (R. 15) (1)

15. You finish your watch and next return to the bridge for the First, the fog has lifted and visibility is in excess of 10nm. You were told in your handover that switchboard drills are taking place. All of a sudden the lights go out, the vent goes silent and the SCC pipes "Total Electrical Failure" and report they have inadvertently tripped all the diesels, you have temporarily lost all propulsion but there is no immediate risk of collision.

a. As you have no power available, what special circumstance criteria do you meet?(R3f) (1)

b. Before coming to rest, what lights should you display?(R27ai&iii) (1)

c. Once the ship is stationary in the water, what lights should be displayed? (R27ai) (1)

Power is restored and you recommence your passage with standard nav lights displayed.

16. Whilst discussing Navigation lights with the YO the subject of submarines comes up. They ask whether or not the submarine must comply with the provisions of the Rules with regard to the display of navigation lights when on the surface. What do the Rules state? (R. 1e) (2)

17. a. A Merchant Vessel is approaching from your Red 30, you are on her stbd side. Risk of collision exists. When two power driven vessels are approaching so as to involve risk of collision what action shall the MV take and what should she avoid? (R.15) (2)

b. The MV seems to be taking little action. When may the stand-on vessel take action? (R17aii) (1)

c. She is still not doing anything, and in fact you can't see anybody on the bridge. When shall the stand-on vessel take such action as will best aid to avoid collision? (R17b) (1)

- d. When taking action as will best aid to avoid collision, is the stand-on vessel constrained when altering her course? If constrained, explain with reference to the Rules. (R17c) (2)

This vessel passes clear and you continue your passage.

18. Ahead, you sight a vessel engaged in fishing, other than trawling, underway and making way. What lights should this vessel display, and if you saw an extra white light, what would this signify? (R. 26) (2)

19. You see ahead on radar a small slow moving contact, you suspect it is a sailing vessel based on its course and speed. What nav lights do you expect it to display and what additional lights may it display. (R. 25a, c) (2)

20. You are about to join the Traffic Separation Scheme (TSS) off the coast of Spain. How should you join or leave a TSS? R(10biii) (2)

21. Having entered the TSS you sight at Red 45, 3nm, a vessel engaged in fishing in the Traffic Separation Zone, we are on her starboard bow. She starts to stray into the Traffic Lane and is closing you on a steady bearing. Who should avoid impeding whom and Why? (R10i) (2)

22. The vessel engaged in fishing continues to close and risk of collision exists. Who is the give way vessel and why? (R8fiii) (2)

ANSWERS

1. Do the Rules apply to the Sea Plane? If so, how should the Sea Plane operate, once it has landed? (R.18e) (3)

Yes, when manoeuvring on the water.(1)

A seaplane on the water shall, in general, keep well clear of all vessels and avoid impeding their navigation.(1) In circumstance, however, where Risk of collision exist, she shall comply with the rules of this part.(1)

5. As you assess there is a risk of collision, does this change any obligations between you and the Sea Plane? Explain your answer (R.8fiii) (2)

No.(1) A vessel required not to impede the passage or safe passage of another vessel is not relieved of this obligation if approaching another vessel so as to involve risk of collision and shall, when taking action, have full regard to the action which may be required by the rules of this part. (1)

The Sea Plane passes at a safe distance, and you continue on passage towards Toulon, to meet with the rest of the exercise group.

3. In determining a safe speed what factors shall be among those taken into account?

- a. By all vessels: (R6) (3)

(i) the state of visibility;

(ii) the traffic density including concentrations of fishing vessels or any other vessels;

(iii) the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;

(iv) at night the presence of background light such as from shore lights or from back scatter of her own lights;

(v) the state of wind, sea and current, and the proximity of navigational hazards;

(vi) the draught in relation to the available depth of water.

(1/2 mark per answer)

- b. And by vessels with operational radar? (R6) (3)

(i) the characteristics, efficiency and limitations of the radar equipment;

(ii) any constraints imposed by the radar range scale in use;

(iii) the effect on radar detection of the sea state, weather and other sources of interference;

(iv) the possibility that small vessels, ice and other floating objects, may not be detected by radar at an adequate range;

(v) the number, location and movement of vessels detected by radar;

(vi) the more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.

(1/2 mark per answer)

4. You have received a Navtex message indicting that there is a 60m tug towing a 200m barge ivo Europa point. You see a double echo on the radar at a range of 8 miles. What light configuration would you expect to see on that bearing if it was the tug and tow coming towards you? (R24, Annex I 2 e) (4)

On the tug: 3 masthead lights in a vertical line(1) with either a single masthead light abaft and higher;(1) or a single forward masthead light with 3 after masthead lights in a vertical line. Sidelights.(1)
On the barge: Sidelights (1)

5. You continue east and you sight visually a container ship on the ships head on a reciprocal course. Who should give way? (R. 14a) (1)

When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other. (1)

6. What do the Rules state about if you have any doubt as to this situation? (R. 14c) (1)

When a vessel is in any doubt as to whether such a situation exists she shall assume that it does exist and act accordingly. (1)

7. How large an alteration should be made to avoid collision? (R. 8b) (1)

It shall be large enough to be readily apparent to another vessel observing visually or by radar.(1)

8. You notice you are slowly overhauling a small merchant vessel on a similar track to your own. When shall you be deemed to be overtaking? (R. 13b) (2)

A vessel shall be deemed to be overtaking when coming up with another vessel from a direction more than 22.5 degrees abaft her beam (1), that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights (1).

9. Who should keep out of the way? (R. 13a) (1)

Notwithstanding anything contained in the Rules of Part B, Sections I and II, any vessel overtaking any other shall keep out of the way of the vessel being overtaken. (1)

10. In construing and complying with these rules due regard shall be had to... (R.2b) (2)

In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances (1), including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger (1).

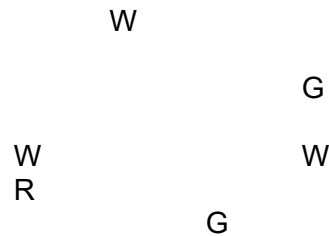
11. Define the term "power-driven vessel" (R.3b) (1)

The term "power-driven vessel" means any vessel propelled by machinery.

12. Define the term "wing in ground craft" (R.3m) (2)

The term “Wing-in-Ground (WIG) craft” means a multimodal craft which (1), in its main operational mode, flies in close proximity to the surface by utilizing surface-effect action(1).

13. You sight the below lights at Green 45 amongst a collection of similar lights.



a. What is this vessel? (R.26 & Annex II) (3)

Vessel engaged in trawling(1), probably over 50 metres in length, underway and making way(1), hauling nets(1).

b. Would any of the vessels lights change if it was not making way through the water? (R.26) (1)

The vessel only shows sidelights when making way through the water. Switch off side and stern lights.

14. A vessel is sighted at RED 50, range 4 miles just visual coming out of the fog bank. The range is closing and the bearing is moving very slowly right. You are 40° on his starboard bow. What action is required by the Rules? (R. 15) (1)

Stand on but monitor the give-way vessel:

(When two power-driven vessels are crossing so as to involve a risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.)

15. You finish your watch and next return to the bridge for the Dogs. You were told in your handover that switchboard drills are taking place. All of a sudden the lights go out, the vent goes silent and the SCC pipes “Total Electrical Failure” and report they have inadvertently tripped all the diesels, you have temporarily lost all propulsion but there is no immediate risk of collision.

a. As you have no power available, what special circumstance criteria do you meet?(R3f) (1)

Not Under Command

b. Before coming to rest, what lights should you display? (R27ai&iii) (1)

Two all-round red lights in a vertical line where they can best be seen(1/2); when making way through the water, ...sidelights and sternlight(1/2).

- c. Once the ship is stationary in the water, what lights should be displayed? (R27ai)
(1)

Two all-round red lights in a vertical line where they can best be seen only.

16. Whilst discussing Navigation lights with the YO the subject of submarines comes up. They ask whether or not the submarine must comply with the provisions of the Rules with regard to the display of navigation lights. What do the Rules state? (R. 1e) (2)

Whenever the Government concerned shall have determined that a vessel of any special construction or purpose cannot comply with the provisions of any of these Rules with respect to the number, position(1/2), range or arc of visibility of lights or shapes(1/2), as well as to the disposition and characteristics of sound-signalling appliances, such vessel shall comply with such other provisions in regard to the number, position, range or arc of visibility of lights or shapes (1/2), as well as to the disposition and characteristics of sound-signalling appliances, as her Government shall have determined to be the closest possible compliance with these Rules in respect of that vessel (1/2).

17. a. A Merchant Vessel is approaching from your Red 30, you are on her stbd side. Risk of collision exists. When two power driven vessels are approaching so as to involve risk of collision what action shall the MV take and what should she avoid? (R.15) (2)

When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her starboard side shall keep out of the way (1) and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel(1).

- b. The MV seems to be taking little action. When may the stand-on vessel take action? (R17aii) (1)

The stand on vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

- c. She is still not doing anything, and in fact you can't see anybody on the bridge. When shall the stand-on vessel take such action as will best aid to avoid collision? (R17b) (1)

When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way vessel alone she shall take such action as will best aid to avoid collision.

- d. When taking action as will best aid to avoid collision, is the stand-on vessel constrained when altering her course? If constrained, explain with reference to the Rules. (R17c) (2)

Yes. (1) She shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side(1).

18. Ahead, you sight a vessel engaged in fishing, other than trawling, underway and making way. What lights should this vessel display, and if you saw an extra white light, what would this signify? (R. 26) (2)

Two all-round lights in a vertical line, the upper being red and the lower white, or a shape consisting of two cones apexes together in a vertical line one above the other; (1/2)

When making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight. (1/2)

When there is outlying gear extending more than 150m horizontally from the vessel, an all-round white light or a cone apex upwards in the direction of the gear; (1)

19. You see ahead on radar a small slow moving contact, you suspect it is a sailing vessel based on its course and speed. What nav lights do you expect it to display and what additional lights may it display. (R. 25a, c) (2)

**A sailing vessel underway shall exhibit:
Side lights and a sternlight(1)**

A sailing vessel underway may, in addition to the lights prescribed in (a) of this rule, exhibit at or near the top of the mast, where they can be seen, two all round lights in a vertical line, the upper being red and the lower being green(1).

20. You are about to join the Traffic Separation Scheme (TSS) off the coast of Spain. How should you join or leave a TSS? R(10biii) (2)

Normally join or leave a TSS at the termination of the lane, (1) but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable(1).

21. Having entered the TSS you sight at Red 45, 3nm, a vessel engaged in fishing in the Traffic Separation Zone, we are on her starboard bow. She starts to stray into the Traffic Lane and is closing you on a steady bearing. Who should avoid impeding whom and why? (R10i) (2)

The fishing vessel should avoid impeding us(1). A vessel engaged in fishing shall not impede the passage of any vessel following a traffic lane(1).

22. The vessel engaged in fishing continues to close and risk of collision exists. Who is the give way vessel and why? (R8fiii) (2)

A vessel the passage of which is not to be impeded remains fully obliged to comply with the rules of this part (1) when the two vessels are approaching one another so as to involve risk of collision. Again we are not to be impeded but risk of collision exists so Rule 18 – Responsibilities between vessels comes into force under Rule 8(f)(iii). Therefore we are the give-way vessel, because ‘a power-driven vessel underway shall keep out the way of a vessel engaged in fishing’(1).